BENCHPILOT EXPERIMENT REPORT

Zinc-air prototype — latest sustained output

Generated from validated structured data
7/18/2026 · demo mode

OBJECTIVE

Determine which construction change raised loaded voltage from approximately 0.46–0.48 V in earlier versions to approximately 1.10 V, and whether the improvement is reproducible.

SETUP

Apparatus: Improvised Tic Tac container cell, Analog Devices Scopy voltmeter, Same or similar electrical load
Materials: Zinc anode (Oxidizing electrode), Air cathode (Oxygen-reduction electrode), Acrylic binder (Cathode binder), Potassium hydroxide electrolyte (Alkaline electrolyte), Stainless-steel mesh support (Cathode support/current collector)

MEASUREMENTS

Start

Fresh open-circuit voltage
Fresh cell at open circuit before the latest loaded reading.

`1.692 V`

Time not recorded

Latest loaded voltage
Under the same or a similar load; elapsed time and load current were not recorded.

`1.1 V`

UNCERTAINTY AND MISSING CONTROLS

Electrical load identity and fan current were not recorded. The latest and older voltage values may reflect different demand.
Resolve: Use one characterized load and log synchronized current.

Elapsed time for the approximately 1.10 V reading is unknown. The point cannot be placed honestly on a voltage-versus-time chart.
Resolve: Use a fixed acquisition schedule beginning at load connection.
Electrolyte quantity, placement, and cathode wetting or flooding were not logged. Ionic wetting may differ between builds.
Resolve: Weigh electrolyte and the assembled cell before each test.
Cathode thickness, mass loading, and binder ratio were not logged. Reaction area and oxygen transport may have changed together.
Resolve: Measure thickness, dry mass, area, and recipe by batch.
Oxygen access, exposed cathode area, and ambient airflow were not controlled. Oxygen availability may explain some or all of the gain.
Resolve: Use a measured air aperture and fixed orientation/airflow.
Contact resistance was not measured. A connection improvement could raise loaded terminal voltage.
Resolve: Use fixed-pressure contacts and measure connection drops.
Clamping pressure was not measured or held constant. Pressure can change both electrical contact and electrode/electrolyte geometry.
Resolve: Use a fixed-pressure fixture and record its setting.
Zinc surface condition and preparation were not logged. Oxide state, cleanliness, and active area may differ between builds.
Resolve: Standardize zinc grade, area, cleaning, and time to assembly.
Cell and ambient temperature were not recorded. Electrochemical kinetics, electrolyte conductivity, and fan demand may vary with temperature.
Resolve: Record temperature and test matched cells in one controlled interval.
Time since cell assembly was not matched or recorded. Wetting, corrosion, and oxygen access may evolve before the load is applied.
Resolve: Use a fixed conditioning time and timestamp assembly and load connection.
The latest result is a single reported run. Reproducibility is unknown and causal attribution is premature.
Resolve: Build and test at least three matched latest-format cells.
Missing controls: Exact load identity, current, and power; Elapsed time of the approximately 1.10 V reading; Electrolyte dose, placement, hydration, and conditioning time; Cathode thickness, mass loading, binder ratio, porosity, and exposed air area; Contact materials, pressure, and resistance; Complete old-versus-latest construction change log; Matched replicate count; Current, internal resistance, and output power cannot be calculated without measured current or load resistance

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COMPETING HYPOTHESES

Electrolyte redistribution / wetting · low confidence. Uneven KOH distribution changes ionic contact area over time. Progressive wetting could lower ionic resistance in the recovering run, while local depletion or poor initial wetting could drive the rapid collapse.
Falsifier: If a controlled electrolyte rewet or redistribution step produces no repeatable change in loaded voltage or internal resistance while connections and airflow remain fixed, this mechanism loses support.
Air-cathode transport state · low confidence. The carbon/MnO_2 cathode may alternate between oxygen starvation and electrolyte flooding. Air-path opening, binder distribution, or gradual gas access could therefore make voltage recover in one assembly but decay in another.
Falsifier: If a controlled change in exposed air area or airflow leaves the load transient unchanged across matched replicates, oxygen transport alone is unlikely to explain the divergence.
Intermittent electrical contact · low confidence. Variable clip pressure, oxide films, or changing mechanical contact could add series resistance. Settling could improve the earlier run; a degrading junction could collapse the current run under load.
Falsifier: If four-wire or fixed-pressure connections show stable contact resistance while the voltage trajectories still diverge, contact variation is not the dominant cause.

RECOMMENDED NEXT TEST

Matched latest-build replication — Before attributing cause, establish whether the approximately 1.10 V loaded result survives matched rebuilds under a characterized load.

Evidence boundary: This report separates user-reported facts, image-derived observations, calculations, hypotheses, and recommendations. It is decision support, not proof, certification, or a substitute for laboratory safety review.